# Computer Science & IT



# Data Structure & Programming

## Graph & Hashing

Lecture No. 03



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# Recap of Previous Lecture



Topic — Hashing

Topic — Open hashing - chaining

Topic — closed Hashing / Open Addressing Linear probing

Topic

Topic

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# Topics to be Covered



- Topic — Quadratic hashing
- Topic — Double Hashing
- Topic
- Topic
- Topic

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# Topic : Hashing

How many different insertion sequences of the key values using the same hash function and linear probing will result in the hash table shown above?

(A) 10

(B) 20

(C) 30

(D) 40

$4! = 24$

42, 23, 34, 46, 52, 33

23, 42, 34, 46, 52, 33

⋮

| | |
|---|---|
| 0 | |
| 1 | |
| 2 | 42 |
| 3 | 23 |
| 4 | 34 |
| 5 | |
| 6 | 46 |
| 7 | |
| 8 | |
| 9 | |

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# Topic : Hashing

How many different insertion sequences of the key values using the same hash function and linear probing will result in the hash table shown above?

(A) 10

(B) 20

(C) 30

(D) 40

42, 23, 34, 52, 46, 33

Any order

3!

42, 34, 23, 52, 46, 33

| | |
|---|---|
| 0 | |
| 1 | |
| 2 | 42 |
| 3 | 23 |
| 4 | 34 |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |

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# Topic : *Primary Clustering*

primary cluster : Long run of filled slot.

once the cluster forms bigger it gets faster it grows

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# Topic : *Primary Clustering*

Primary clustering is the tendency for a collision resolution scheme such as linear probing to create *long runs of filled slots* near the hash position of keys.

If the primary hash index is $x$, subsequent probes go to $x + 1, x + 2, x + 3$ and so on, this results in Primary Clustering.

Once the *primary cluster forms*, the bigger the cluster gets, the *faster it grows*. And it *reduces the performance*.



# Topic : Hashing

Quadratic probing




probe$_0$ = h(x)

probe$_1$ = (h(x) + $1^2$) % m

probe$_2$ = (h(x) + $2^2$) % m

probe$_3$ = (h(x) + $3^2$) % m

probe$_4$ = (h(x) + $4^2$) % m

⋮

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# Example

{96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166}.

| $Key - x$ | $h(x,i) = (h(x), i^2) \ mod \ 20$ |
|---|---|
| 96 | 16 |
| 48 | 8 |
| 63 | 3 |
| 29 | 9 |
| 87 | 7 |
| 77 | 17 |
| 47 | 7 |
| 73 | |
| 70 | |
| 86 | |
| 166 | |

47%20 = 7
occupied

probe1 = 47%20 + $1^2$
= 7 + 1 = 8

probe2 = 47%20 + $2^2$
= 7 + 4 = 11

| | |
|---|---|
| | 0 |
| | 1 |
| | 2 |
| 63 | 3 |
| | 4 |
| | 5 |
| | 6 |
| 87 | 7 |
| 48 | 8 |
| 29 | 9 |
| | 10 |
| 47 | 11 |
| | 12 |
| | 13 |
| | 14 |
| | 15 |
| 96 | 16 |
| 77 | 17 |
| | 18 |
| | 19 |

# Example

{96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166}.

| $Key - x$ | $h(x,i) = (h(x), i^2) \ mod \ 20$ |
|---|---|
| 96 | 16 |
| 48 | 8 |
| 63 | 3 |
| 29 | 9 |
| 87 | 7 |
| 77 | 17 |
| 47 | 7 |
| 73 | 13 |
| 70 | 10 |
| 86 | 6 |
| 166 | |

$166\%20 = 6$

occupied

probe 1

$166\%20 + 1^2$

$= 7$

probe 2

$166\%20 + 2^2 =$

$6 + 4 = 10$

probe 3

$166\%20 + 3^2$

$6 + 9 = 15$

| | |
|---|---|
| | 0 |
| | 1 |
| | 2 |
| 63 | 3 |
| | 4 |
| | 5 |
| 86 | 6 |
| 87 | 7 |
| 48 | 8 |
| 29 | 9 |
| 70 | 10 |
| 47 | 11 |
| | 12 |
| 73 | 13 |
| | 14 |
| 166 | 15 |
| 96 | 16 |
| 77 | 17 |
| | 18 |
| | 19 |

# Topic : Hashing

| Key | $h(x,i) = (h(x), i^2) \ mod \ 10$ |
|---|---|
| 2 | 2 |
| 3 | 3 |
| 6 | 6 |
| 8 | 8 |
| 7 | 7 |
| 11 | 1 |
| 12 | |
| | |
| | |
| | |
| | |

probe$_0$ = 2 = h(k)

probe$_1$ = 2 + $1^2$ = 3

probe$_2$ = 2 + $2^2$ = 6

probe$_3$ = (2 + $3^2$) % 10
= 1

probe$_4$ = (2 + $4^2$) % 10
= 8

probe$_5$ = (2 + $5^2$) % 10
= 7

probe$_6$ = (2 + $6^2$) % 10
= 8

| | |
|---|---|
| | 0 |
| 11 | 1 |
| 2 | 2 |
| 3 | 3 |
| | 4 |
| | 5 |
| 6 | 6 |
| 7 | 7 |
| 8 | 8 |
| | 9 |

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# Topic : Hashing

In quadratic probing some key value may cycle through the list

e.g. 12 will not Inserted



# Topic : Hashing

## Double Hashing

In Double Hashing there are two hash function

1. primary hash function [probe0]
2. Secondary hash function [offset]

Linear probing offset is always 1

In Double Hashing offset will not be fixed

Its calculated by Secondary Hash function

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# Topic : Hashing

probe = $h_1(k)$;

offset = $h_2(k)$

repeat the following until empty slot found

probe = (probe + offset) % m

k = 44

probe = 5

offset = 8 - (x % 8)

= 8 - (44 % 8)

= 4

probe1 = (5 + 4) % 13 = 9

probe2 = (9 + 4) % 13 = 0

# Topic : Hashing

$h_1(x) = x \% 13$  $h_2(x) = 8 - (x \% 8)$

| Key – x | $h_1(x) = x \bmod 13$ | $h_2(x) = 8 - (x \bmod 8)$ |
|---|---|---|
| 18 | 5 | |
| 41 | 2 | |
| 22 | 9 | |
| 44 | 5 | 8 – (44 % 8) = 4 |
| 59 | | |
| 32 | | |
| 31 | | |
| 73 | | |

| | |
|---|---|
| 44 | 0 |
| | 1 |
| 41 | 2 |
| | 3 |
| | 4 |
| 18 | 5 |
| | 6 |
| | 7 |
| | 8 |
| 22 | 9 |
| | 10 |
| | 11 |
| | 12 |

probe1
= (5+4) % 13
= 9

probe2
(9+4) % 13
= 0

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# Topic : Hashing

$h_1(x) = x \% 13$  $h_2(x) = 8 - (x \% 8)$

| Key – x | $h_1(x) = x \bmod 13$ | $h_2(x) = 8 - (x \bmod 8)$ |
|---|---|---|
| 18 | 5 | |
| 41 | 2 | |
| 22 | 9 | |
| 44 | 5 | 8 – (44%8) = 4 |
| 59 | 7 | |
| 32 | 6 | |
| 31 | 5 | 8 – (31%8) = 8-7 = 1 |
| 73 | | |




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# Topic : Hashing

$h_1(x) = x \% 13$  $h_2(x) = 8 - (x \% 8)$

| Key – x | $h_1(x) = x \bmod 13$ | $h_2(x) = 8 - (x \bmod 8)$ |
|---|---|---|
| 18 | 5 | |
| 41 | 2 | |
| 22 | 9 | |
| 44 | 5 | $8-(44\%8)=4$ |
| 59 | 7 | |
| 32 | 6 | |
| 31 | 5 | $8-(31\%8)=8-7=1$ |
| 73 | 8 | $8-(73\%8)=7$ |

| | |
|---|---|
| 44 | 0 |
| | 1 |
| 41 | 2 |
| 73 | 3 |
| | 4 |
| 18 | 5 |
| 32 | 6 |
| 59 | 7 |
| 31 | 8 |
| 22 | 9 |
| | 10 |
| | 11 |
| | 12 |

probe = 8

$probe_1 = (8+7)\%13 = 15\%13 = 2$

$probe_2 = (2+7)\%13 = 9$

$probe_3 = (9+7)\%13 = 3$

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# Topic : Hashing

Consider a double hashing scheme in which the primary has function is h, (k) = k mod 23, and the secondary hash function is h, (k) =1+(k mod 19). Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value k = 90 is

$h_1(k) = k \bmod 23$

$h_2(k) = 1 + (k \bmod 19)$

probe0 = 90 % 23

= 21

offset = 1 + (90 % 19)

= 15.

probe0 = (21 + 15) % 23

= 36 % 23 = 13

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# Topic : Hashing

Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 10$, and the secondary hash function is $h_2(k) = 7 - (k \bmod 7)$. Assume that the table size is 10. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k = 49$ is

# Topic : Hashing

2025 Set-1

#Q. In a double hashing scheme, $h1(k)=k \bmod 11$ and $h2(k)=1+(k \bmod 7)$ are the auxiliary hash functions. The size $m$ of the hash table is 11. The hash function for the i-th probe in the open address table is $[h1(k)+i\ h2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is [10]. (Answer in integer)

$h_1(k) = k \% 11$

$h_2(k) = 1 + (k \% 7)$

$[h_1(k) + i * h_2(k)] \% m$




probe : 24 % 11 = 2

offset : 1 + (24 % 7)
= 1 + 3 = 4

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# 2 mins Summary

Topic — Quadratic probing

Topic — Double Hashing

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# THANK - YOU

June - July

practice + Revision

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